

Atividade 04 - Modelagem e Resolução pelo Método Simplex

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26

Dom Seg Ter Qua Qui Sex Sáb

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1. $\max Z = 400x_1 + 700x_2$
 sujeito a

$$\begin{aligned}
 2x_1 + 3x_2 &\leq 180 & 2x_1 + 3x_2 + s_1 &= 180 \\
 x_1 + 3x_2 &\leq 150 & x_1 + 3x_2 + s_2 &= 150 \\
 2x_1 + 4x_2 &\leq 200 & 2x_1 + 4x_2 + s_3 &= 200 \\
 x_2 &\leq 40 & x_2 + s_4 &= 40 \\
 x_1, x_2 &\geq 0
 \end{aligned}$$

A base inicial é $\{s_1, s_2, s_3, s_4\}$. Fazendo $x_1 = x_2 = 0$:

$SBV = (x_1, x_2, s_1, s_2, s_3, s_4) = (0, 0, 180, 150, 200, 40)$

Base	x_1	x_2	s_1	s_2	s_3	s_4	b	
s_1	2	3	1	0	0	0	180	$180/3=60 \quad L_1=L_1-3L_4$
s_2	1	3	0	1	0	0	150	$150/3=50 \quad L_2=L_2-3L_4$
s_3	2	4	0	0	1	0	200	$200/4=50 \quad L_3=L_3-4L_4$
s_4	0	1	0	0	0	1	40	✓
Z	-400	-700	0	0	0	0	0	$L_z=L_z+700L_4$

Linha	x_1	x_2	s_1	s_2	s_3	s_4	b
L_1	2	0	1	0	0	-3	60
L_2	1	0	0	1	0	-3	30
L_3	2	0	0	0	1	-4	40
L_4	0	1	0	0	0	1	40
L_z	-400	0	0	0	0	700	28000



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Base	x_1	x_2	s_1	s_2	s_3	s_4	b
s_1	2	0	1	0	0	-3	60 $60/2=30$
s_2	1	0	0	1	0	-3	30 $30/1=30$
s_3	2	0	0	0	1	-4	40 $40/2=20$ $L_3=L_3/2$
x_2	0	1	0	0	0	1	40
Z	-400	0	0	0	0	700	28000

Linha	x_1	x_2	s_1	s_2	s_3	s_4	b
L_1	2	0	1	0	0	-3	60 $L_1=L_1-2L_3$
L_2	1	0	0	1	0	-3	30 $L_2=L_2-L_3$
L_3	1	0	0	0	$1/2$	-2	20
L_4	0	1	0	0	0	1	40
L_2	-400	0	0	0	0	700	28000 $L_2=L_2+400L_3$

Base	x_1	x_2	s_1	s_2	s_3	s_4	b
s_1	0	0	1	0	-1	1	20 $\checkmark$
s_2	0	0	0	1	$-1/2$	-1	10 $L_2=L_2+L_1$
x_1	1	0	0	0	$1/2$	-2	20 $L_3=L_3+2L_1$
x_2	0	1	0	0	0	1	40 $L_4=L_4-L_1$
Z	0	0	0	0	200	-700	36000 $L_2=L_2+100L_1$

Base	x_1	x_2	s_1	s_2	s_3	s_4	b	Conferência
s_4	0	0	1	0	-1	1	20	$2(60)+3(20) \leq 280$
s_2	0	0	1	1	$3/2$	0	30	$60+3(20) \leq 150$
x_1	2	0	2	0	$3/2$	0	60	$2(60)+4(20) \leq 200$
x_2	0	1	-1	0	1	0	20	$20 \leq 40$
Z	0	0	200	0	100	0	38000	

$$400(60) + 700(20) = 38000$$





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$$2. \max Z = 600x_1 + 500x_2$$

$$\begin{array}{ll} \text{sujeito a} & 2x_1 + x_2 \leq 100 \\ & x_1 + 2x_2 \leq 80 \\ & 3x_1 + x_2 \leq 120 \\ & x_1 \leq 45 \\ & x_1, x_2 \geq 0 \end{array} \quad \begin{array}{l} 2x_1 + x_2 + s_1 = 100 \\ x_1 + 2x_2 + s_2 = 80 \\ 3x_1 + x_2 + s_3 = 120 \\ x_1 + s_4 = 45 \end{array}$$

A base inicial é $\{s_1, s_2, s_3, s_4\}$, fazendo $x_1 = x_2 = 0$:

$$SBV = (x_1, x_2, s_1, s_2, s_3, s_4) = (0, 0, 100, 80, 120, 45)$$

Base	x_1	x_2	s_1	s_2	s_3	s_4	b
s_1	2	1	1	0	0	0	100 ⁵⁰
s_2	1	2	0	1	0	0	80 ⁴⁰
s_3	3	1	0	0	1	0	120 ⁴⁰ $\checkmark L_3 = L_3/3$
s_4	1	0	0	0	0	1	45 ⁴⁵
Z	-600	-500	0	0	0	0	0

Linha	x_1	x_2	s_1	s_2	s_3	s_4	b
L_1	2	1	1	0	0	0	100 $L_1 = L_1 - 2L_3$
L_2	1	2	0	1	0	0	80 $L_2 = L_2 - L_3$
L_3	1	$1/3$	0	0	$1/3$	0	40
L_4	1	0	0	0	0	1	45 $L_4 = L_4 - L_3$
L_2	-600	-500	0	0	0	0	0 $L_2 = L_2 + 600L_3$



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Base	x_1	x_2	s_1	s_2	s_3	s_4	b
s_1	0	$\frac{2}{3}$	1	0	$-\frac{2}{3}$	0	20 $\hookrightarrow$
s_2	0	$\frac{5}{3}$	0	1	$-\frac{1}{3}$	0	40 $\xrightarrow{24} \sqrt{L_2 = \frac{3}{5}L_2}$
x_1	1	$\frac{2}{3}$	0	0	$\frac{2}{3}$	0	40 $\xrightarrow{100}$
s_4	0	$-\frac{2}{3}$	0	0	$\frac{1}{3}$	1	5
Z	0	-300	0	0	200	0	24000

linha	x_1	x_2	s_1	s_2	s_3	s_4	b
L_1	0	$\frac{2}{3}$	1	0	$-\frac{2}{3}$	0	20 $L_1 = L_1 - \frac{2}{3}L_2$
L_2	0	1	0	$\frac{3}{5}$	$-\frac{1}{5}$	0	24
L_3	1	$\frac{2}{3}$	0	0	$\frac{2}{3}$	0	40 $L_3 = L_3 - \frac{2}{3}L_2$
L_4	0	$-\frac{2}{3}$	0	0	$\frac{1}{3}$	1	5 $L_4 = L_4 + \frac{2}{3}L_2$
L2	0	-300	0	0	200	0	24000

Base	x_1	x_2	s_1	s_2	s_3	s_4	b
s_1	0	0	1	$-\frac{2}{5}$	$-\frac{3}{5}$	0	12 $x_1 = 32$
x_2	0	1	0	$\frac{3}{5}$	$-\frac{1}{5}$	0	24 $x_2 = 24$
x_1	1	0	0	$-\frac{2}{5}$	$\frac{2}{5}$	0	32
s_4	0	0	0	$\frac{1}{5}$	$-\frac{2}{5}$	1	13
Z	0	0	0	180	140	0	312000

Conferência

$$2(32) + 24 \leq 100$$

$$32 + 2(24) \leq 80$$

$$3(32) + 24 \leq 120$$

$$32 \leq 45$$

$$600(32) + 500(24) = 31200$$

